

WAVSYNTH PARAMETERS	
SHAPE 00—03	Pulse 12%, 25%, 50%, 75% <i>Two-level waves: clean even at 8 bits</i>
SHAPE 04—08	Saw, triangle, sine, pitched noise, noise <i>09 and up are wavetables</i>
SIZE	Waveform length <i>Also sets how smooth PWM sounds</i>
MULT	Repeats the shape <i>Sounds like hard sync as it rises</i>
WARP	Skews the shape <i>Pushes it to one side of the table</i>
SCAN	Mirror or scan <i>Shapes 00—08: mirror point (PWM); wavetables: 64 waves</i>
WAV LP/HP	8-bit filters <i>WAV LP, HP, BP, BS reshape the wave itself</i>
FX	OSC · SIZ · MUL · WRP · SCN <i>Automate each control per step</i>

WAVETABLES 09—45	
09—1A	OSC <i>CRUSH, FOLDING, FUZZY, GHOST, LIQUID, TUBE, WAVES...</i>
1B—21	BNK <i>DRIP, FROGGY, INSONIC, RADIUS, SCRATCH, SMOOTH, WOBBLE</i>
22—2E	HRM <i>GENTLE, HARMONIC, PLAITS01—02, TONAL, TWINE...</i>
2F—43	EFX <i>ALIEN, FORMANT, RINGS, SHIMMER, SPOOKY, VOCAL, ZAP...</i>
44—45	VOX <i>BRAIDS, VOXSYNTH</i>

GOOD TO KNOW	
8-Bit Character	256 LEVELS
- Wavsynth uses 256 levels, so saws look like staircases - Those edges add bright, small-speaker harmonics - Pulse waves have only two levels and stay clean	
Same Tables in FM	FM W SHAPES
- FM Synth operators can use these wavetables too - There, FB scans the table instead of adding feedback	

WAVSYNTH RECIPES	
Chip Pulse Bass	PULSE 50%
- SHAPE PULSE 50%: only two levels, so 8-bit costs nothing - Short AHD on volume; WARP to change its hollow tone - A little MFX send (chorus) widens it	
PWM Lead	SCAN
- SHAPE PULSE 50%, then an LFO on SCAN sweeps the width - Larger SIZE makes the sweep smoother	
Chip Chords	ARP 47
- PULSE 25% with ARP 47 (major) or ARP 37 (minor) - ARC sets the arpeggio mode and speed in ticks - PVB after the attack for a classic wobbly lead	
Noise Drums	NOISE
- SHAPE NOISE with a very short AHD for hats and snares - PITCHED NOISE plays in tune: the note sets its color - Fast PBN down on pitched noise for laser zaps	
Wavetable Morph	SCN
- Pick a wavetable (e.g. 26 HRM:HARMONIC or 3F EFX:VOCAL) - An LFO or table steps on SCAN morph through its 64 waves	
Sync-Style Sweeps	MULT
- AHD on MULT stacks repeats of the shape as it rises - Sounds like a hard-sync sweep on leads and basses	
Tame or Crunch	WAV LP · CUTOFF
8-bit saws step like a staircase and add bright edges - A normal low-pass tames them; WAV filters add crunch	
Duty-Cycle Switching	OSC 00—03
- OSC on a step switches between the four pulse widths 12% is thin and nasal, 50% is hollow and round	
Chip Kick	SINE · AHD PITCH
- SHAPE SINE with a short AHD on volume - A fast AHD on pitch drops it for the thump	
Echo Channel	DEL · V
- Copy the lead phrase to a free track with lower V values - DEL on its first row delays it for a classic tracker echo	

HYPERSYNTH PARAMETERS	
SHAPE	12 shapes (6.6) <i>Saw variants, soft square, triangle, sine variants</i>
CHORD	16 banks, up to 6 intervals <i>SHIFT + OPT copies; SHIFT + EDIT pastes</i>
SCALE	Song scale for the chord <i>00 follows the song or track scale</i>
SHIFT	Crossfade <i>Between intervals 1—3 and 4—6</i>
SWARM	Detune <i>Spreads each note's oscillator pair</i>
WIDTH	Stereo spread <i>00 mono to FF full stereo</i>
SUBOSC	Square sub <i>Below 80: two octaves down · above 80: one</i>
FX	CRD · SWM · WID · SUB <i>Change chords and tone per step</i>

HYPERSYNTH RECIPES	
Lush Pad	SWARM · WIDTH
- CHORD: major 7th = 0, 4, 7, 11 semitones (decimal) - Moderate SWARM, WIDTH toward FF, MFX and REV sends - ADSR on volume, slow ATK and long REL; end notes with OFF	
Chord Progressions	CRD
- Build a bank for each chord shape you need - CRD on a step switches banks; the note sets the root	
Evolving Extensions	SHIFT
- Intervals 1—3: a triad · 4—6: its 7th, 9th and 11th - LFO on SHIFT crossfades the groups; the first fades out	
Supersaw Stab	SWARM · AHD
- SHAPE SAW, high SWARM, WIDTH FF; short volume AHD - A DEL send for trance-style echoes	
Bass + Chord in One	SUBOSC
- Raise SUBOSC for the root under the chord - Low CUTOFF and a short AHD on it for a stab	
Stay in Key	SCALE
- Set SCALE to a song scale so intervals follow the key - SCA or SCG changes the scale mid-song	
Solid Low End	WIDTH
- Keep WIDTH low when SUBOSC or low notes carry the bass - Save full width for pads that sit above it	
Pluck Chords	CUTOFF · AHD
- Low SWARM, short AHD on CUTOFF with some RES - Play off-beat chords for house and garage stabs	